

Buy Now, Pay Later: Handy, or Predatory?

Identifying financial vulnerabilities among users of buy now, pay later services

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Abstract

“Buy now, pay later” (BNPL) services have become increasingly popular since their emergence in the mid-2010’s. Since then, an emerging body of literature has identified exploitative tendencies in the marketing and use of BNPL services, with many large providers catering towards high-risk, low-income, and otherwise vulnerable demographics. We explored several data-driven approaches to identify individual characteristics which would motivate a consumer to opt into a BNPL service, and to identify which users of BNPL services would default on a repayment. Key predictors of BNPL adoption and default on a BNPL loan included one’s access to liquid assets and one’s ability to take out loans and apply for credit cards. In other words, individuals with unstable finances were the most likely to use BNPL services, and to suffer financially for using a BNPL service.

Keywords: BNPL, personal finance, present bias

1. Introduction

“Buy Now, Pay Later” (BNPL for short) refers to repayment programs offered by retailers which allow customers to pay for a purchase in small installments over time. Unlike traditional loan services, BNPL services typically offer smaller loans with zero percent interest rates, which makes using a BNPL service attractive for financing smaller purchases, such as food, clothing, and household goods. Additionally, these services are often built directly onto online purchasing platforms, allowing users to opt in immediately before a purchase. Use of BNPL services increased almost tenfold between 2019 and 2021, indicating increased reliance on these services in the wake of the Covid-19 pandemic (2).

The seamless user experience and lack of cognitive barriers (e.g., having to sign a credit card statement before having a purchase greenlit) is conducive to short-term, impulse-driven purchasing and positive emotions (4). However, these feelings are often followed by regret of

the purchase, anxiety anticipating the next payment, and negative self-evaluation, such as feeling that one cannot trust their own financial decision making. This mistrust is then directed outwards towards the retailer and the provider, potentially creating strain between the consumer and the retailer (4). Despite this, many users of BNPL services misconstrue them as a tool which saves money, rather than a tool which facilitates spending (5).

On paper, that BNPL services are a means of saving should be true; the ability to repay in installments should ease financial burdens on consumers with liquidity restraints, allowing users to purchase “low-ticket” necessities immediately, and pay without interest once they have the means to do so. However, the adoption of BNPL services seems to encourage the opposite behavior in users, in what’s been deemed a “flypaper effect” (3); first-time use of a BNPL service correlates with an increase in spending, more frequent overdrafts, the accessing of long-term savings, and accumulation of late fees. Opening with a small initial fee and allowing the user to divide their payments into smaller increments also tends to increase an individual consumer’s overall spending levels (1).

The user base of BNPL services has come to rely on them more for necessities like groceries, rather than big-ticket luxuries. Over half of users report paying late on a BNPL loan, and one fourth report taking out loans for multiple purchases simultaneously (6). BNPL use overall is associated with economically disadvantaged social groups, and individuals who lack financial resources. Black and hispanic consumers are more likely to use BNPL services than white consumers, women are more likely to borrow than men, and low-income consumers and users without savings are more likely to borrow than high-income consumers. Additionally, younger consumers, presumably lacking intergenerational financial support, are more likely to use BNPL services than older consumers (2).

We propose that the adoption of BNPL services presents a potential hazard to consumers, gradually entrapping them in cycles of debt and repayment. The primary objective of the present study is to predict which individuals in the general population are most “at risk” for adoption of BNPL services, and to predict who among BNPL users is at risk for making a late payment. Additionally, as a secondary objective, we would like to describe demographic features of BNPL users that are yet unrecognized in the literature.

2. Data Collection

We utilized the Survey of Household Economics and Decisionmaking (SHED), a nationwide survey of consumers collected and published annually by the Board of Governors of the Federal Reserve System. Survey subjects include categories such as financial well-being, employment, income, education, and retirement. We selected data collected from 2021 through 2024, which spans the recovery from the COVID-19 pandemic and subsequent inflationary economic pressure. This analysis focuses specifically on the variables BNPL1 (usage of BNPL services in the past 12 months) and BNPL3 (frequency of missed or late payments on these services), which we treat as primary target variables to assess consumer credit risk and financial distress.

3. Methods

3.1. Feature selection (PCA)

Due to the high dimensionality and inherent multicollinearity of the SHED dataset, we utilized Principal Component Analysis (PCA) to refine our feature space. For the BNPL adoption model (BNPL1), we identified 15 principal components that accounted for the majority of the variance associated with consumer adoption. These components were primarily driven by demographic and sentiment-based features, including age, income, marital status, and self-reported financial concerns. In the BNPL delinquency model (BNPL3), the feature set largely overlapped with the usage model; however, additional predictors surfaced as significant, most notably carrying unpaid bill balance frequency. By utilizing these reduced feature sets in our subsequent modeling, we effectively enhanced the models' predictive accuracy while mitigating the risk of overfitting, ensuring the results remain good generalization.

3.2. Descriptive Analysis (EDA)

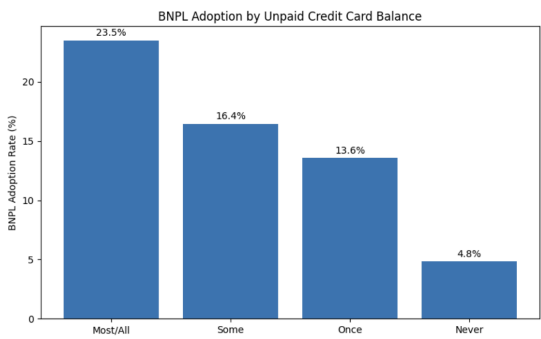


Fig. 1. BNPL adoption by frequency of unpaid credit card balance

We first examined the BNPL adoption rate by the frequency of carrying unpaid credit card balance. Firstly, respondents who frequently carried unpaid credit card balances have significantly higher BNPL adoption rates (23.5%, compared to only 4.8% for non-adopters) (Fig. 1). This suggests that BNPL usage is strongly associated with existing financial stress and

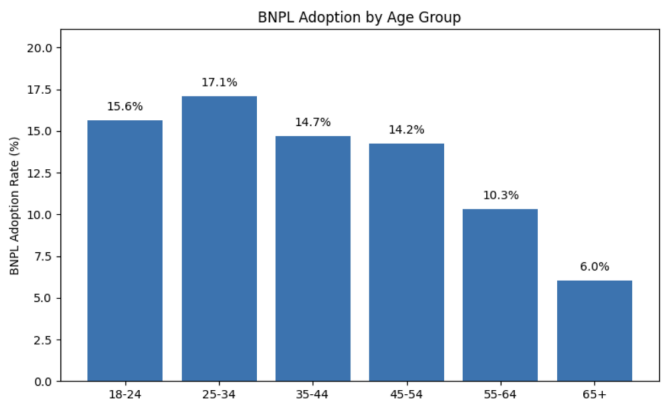


Fig. 2. BNPL adoption rate by age group

short-term liquidity constraints, rather than simply being a matter of payment preference.

BNPL adoption also varied by age. Respondents aged 25–34 had the highest adoption rate at 17.1%, followed by respondents aged 18–24 at 15.6%. In contrast, respondents aged 65 and older had the lowest adoption rate at 6.0% (Fig. 2). This pattern suggests that BNPL use is more common among younger consumers, which is consistent with prior research showing that younger users are more likely to rely on alternative credit tools.

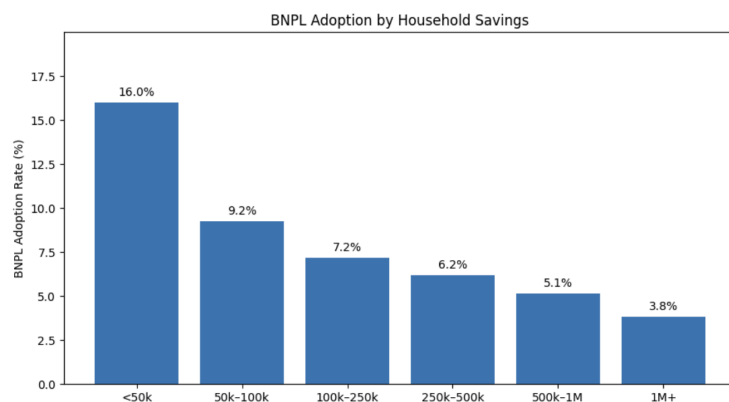


Fig. 3. BNPL adoption rate by household savings

BNPL adoption also showed a clear negative relationship with household savings. Respondents with less than \$50,000 in savings had the highest BNPL adoption rate at 16.0%, while those with more than \$1 million in savings had the lowest rate at only 3.8% (Fig. 3). This supports the argument that BNPL usage is often driven by short-term liquidity constraints rather than convenience alone.

BNPL users are more likely to be younger, financially constrained, and already managing existing credit pressure. Higher adoption rates among individuals with unpaid credit card balances and lower household savings indicate that BNPL is often used as a short-term financial coping tool rather than a simple payment preference. These findings are consistent with both the machine learning models and prior literature on BNPL behavior.

Late payment risk among BNPL users showed a strong relationship with income level (Fig. 4). Respondents earning less than \$10,000 had the highest late payment rate at 33.5%, followed by those earning \$10,000–\$24,999 at 32.1%. In contrast, users earning \$150,000 or more had the lowest late payment rate at only 8.2%. This suggests that lower-income users are much more likely to miss BNPL payments, likely because they rely more heavily on short-term installment credit to manage daily expenses. This finding is highly consistent with our SVM, XGBoost, and SHAP results, where income level (ppinc7) was repeatedly identified as one of the strongest predictors of BNPL default risk.

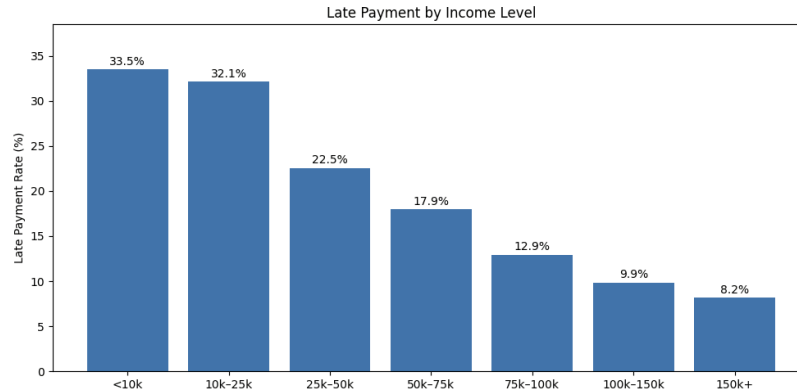


Fig. 4. BNPL delinquency rate by Income Level

Late payment behavior also showed a strong relationship with confidence in getting approved for a credit card. BNPL users who were not confident about credit card approval had the highest late payment rate at 35.6%, followed by those who answered “Don’t know” at 24.7%. In contrast, users who were very confident had the lowest delinquency rate at only 6.6% ([Fig. 5](#)). This suggests that users who cannot easily rely on mainstream credit products may depend more heavily on BNPL services and therefore face greater repayment pressure. This finding is highly consistent with the machine learning results, where A6 was repeatedly identified as one of the strongest predictors of BNPL default.

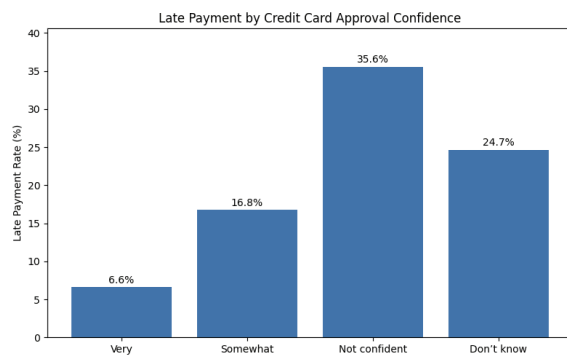


Fig. 5. BNPL delinquency by credit card approval confidence

Late payment rates also differed by household savings among BNPL users. Users with less than \$50,000 in savings had the highest late payment rate at 18.5%, while users with \$1 million or more in savings had the lowest rate at 3.9% ([Fig. 6](#)). This suggests that lower household savings are not only associated with higher BNPL adoption, but also with greater repayment risk after adoption. In other words, savings appear to matter at both stages: whether someone uses BNPL and whether they can repay it on time.

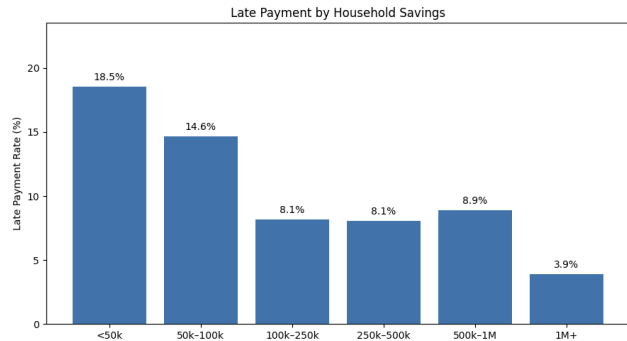


Fig. 6. BNPL delinquency by household savings

Overall, the descriptive analysis in this section shows that BNPL default risk is strongly concentrated among financially vulnerable users. Lower income, weaker access to credit, and lower savings are all associated with a significantly higher probability of late payment. These patterns are highly consistent with the machine learning results from SVM, XGBoost, and SHAP analysis, where income level, credit approval confidence, and savings repeatedly appeared as the most important predictors of BNPL default.

Early exploratory data analysis revealed an imbalance in gender among BNPL users, with a greater proportion of female to male users. We speculated the higher representation of women would be related to motherhood, and thus included variables such as household size, number of children, and marital status in our analysis.

We confirmed that women with children were approximately 80% more likely to adopt BNPL services than women without children ([Fig. 7](#)). Mothers with multiple children were increasingly likely to use a BNPL service depending on the number of children in their household ([Fig. 8](#)). These effects were statistically significant until the sixth child, as the number of families with six children in our dataset was small ($n=55$). This effect persisted regardless of income level ([Fig. 9](#)) and marital status ([Fig. 10](#)), indicating a strong relationship specifically between having children, and dependency on a BNPL service. Factoring in the increasing use of BNPL services as a means of securing groceries and other necessities, we interpret this to mean that mothers should be considered one of the financially challenged demographics who are susceptible to BNPL adoption.

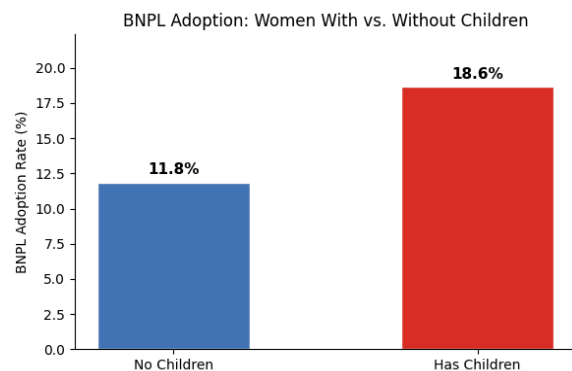


Fig. 7. BNPL adoption rate by presence of children among women

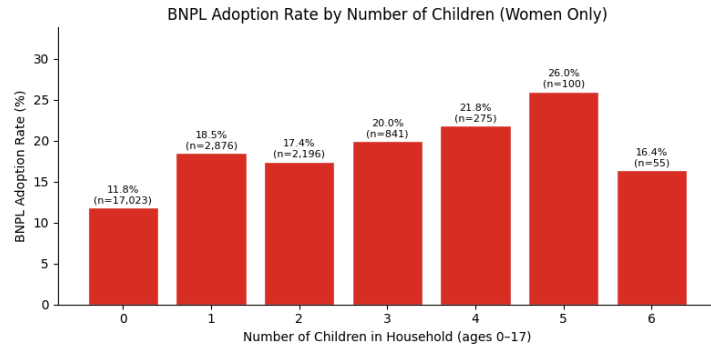


Fig. 8. BNPL adoption rate by number of children in household

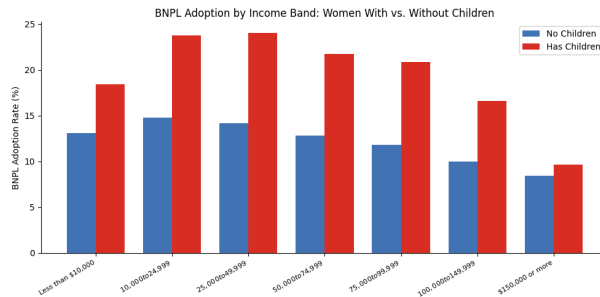


Fig. 9. BNPL adoption rate by household income & presence of children

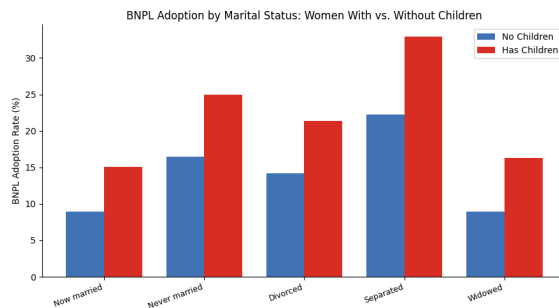


Fig. 10. BNPL adoption rate by marital status & presence of children

3.3. Modeling pipeline

The modeling pipeline was engineered to maximize predictive performance while maintaining a rigorous focus on generalization and the prevention of overfitting through systematic validation and regularization.

3.3.1. Data Preprocessing and Engineering: The raw data underwent a comprehensive pre-processing. Binary variables were mapped to a 0/1 indicator, while multi-class categorical variables were processed via One-Hot Encoding. To ensure that no information from the test set influenced the training phase, we applied a StandardScaler to normalize the predictive variable values, fitting the scaler on the train, test and validation set separately. To ensure the findings are statistically representative of the national adult population, all models integrated weights provided in the SHED dataset.

3.3.2. Oversampling: To address the significant class imbalance between BNPL users and non-users, we implemented Random Oversampling (ROS) on the training dataset. By oversampling minority class records, this technique ensures that the model puts equal weight to both outcomes, preventing a predictive bias.

2.3.3. Model Selection and Validation Strategy: We splitted data into Training, Validation, and Test sets—and utilized k-fold cross-validation. This approach allowed us to systematically monitor validation loss against training loss, ensuring that the models captured generalizable patterns rather than noise.

3.3.4. Optimization: To maximize predictive performance, we conducted extensive Hyperparameter Tuning for high-capacity models. For XGBoost, we optimized parameters such as the learning rate, maximum tree depth, and subsample ratio. For the Neural Network, we tuned the number of hidden layers, patience etc. We also introduced early stopping to maximize generalization.

3.3.5. Regularization: To mitigate the risk of overfitting, particularly given the high-dimensional nature of the survey components, we introduced L1 (Lasso) and L2 (Ridge) penalties. This regularization strategy, combined with early stopping for the neural network, ensured that our models maintained a high degree of reproductive stability.

3.4. ML algorithms

To identify the most robust predictive framework for BNPL behavior, we benchmarked a diverse suite of algorithms: Logistic Regression, Support Vector Machines (SVM), XGBoost, and Artificial Neural Networks (ANN).

3.4.1. Ordered Logistic Regression: Built as a baseline probabilistic model to establish a linear decision boundary and provide initial insights into feature importance.

3.4.2. Support Vector Machines (SVM): Utilized to capture potential non-linear separations in multi-dimensional consumer profiles.

3.4.3. XGBoost: Implemented as a high-performance gradient-boosted decision tree framework, this model was selected for its interpretability through feature importance rankings, offering insights into the underlying decision logic and the primary consumer characteristics that lead the model to its final predictions.

3.4.4. Artificial Neural Networks (ANN): Developed a Multi-Layer Perceptron (MLP) architecture to move beyond traditional modeling and explore deep-learning-based complex data patterns. The ANN provides a sophisticated benchmark to determine if "hidden" structures in consumer behavior offer additional predictive lift.

The performance of these models was evaluated integrating cross-validation strategy, with a primary focus on metrics such as Loss, F1-Score, AUC-ROC, Precision, and Recall. Given the goal of the research, we placed more weight on minimizing false negatives on minority class (Class 1) in order to ensure the model's capability to detect segments of population who are most engaged with alternative credit products, especially those with potential financial distress which are critical in risk-assessment.

4. Results

4.1. Model Performance Comparison

The predictive performance for both BNPL Adoption (BNPL1) and BNPL Delinquency (BNPL3) was evaluated across four architectures: ANN, Logistic Regression, SVC, and XGBoost. The results are summarized in ([Table. 1](#)). For Adoption, SVC achieved the highest ROC-AUC (0.78), while Logistic Regression, XGBoost, SVC and ANN ([Fig. 11](#)) proved similar efficiency at identifying actual users (Recalls of 0.69, 0.62, 0.68 and 0.62 respectively). All models exhibited high Class 0 Precision (0.91+) with SVC modeling delivering the best performance, demonstrating models' efficiency in identifying marginal customers who do not utilize BNPL services.

In the Default prediction, SVC maintained a leading ROC-AUC (0.77), whereas XGBoost dropped to 0.67. However, XGBoost is the clear leader in Class 1 Recall (0.71) ([Fig. 12](#)), successfully flagging the highest percentage of defaults. The Logistic and SVC and ANN models follow at 0.66, 0.65 and 0.60 respectively, demonstrating a lower ability to identify at-risk users than XGBoost. Furthermore, it maintains a Class 1 Precision less than 0.40, a performance level that remains remarkably consistent across all architectures. This indicates models struggle to isolate actual users without catching a lot of non-users.

The Class 0 Stability: All models demonstrate high Class 0 Precision (0.90–0.95), meaning when a model predicts a "Non-User" or "Non-Defaulter," it is correct nearly 9 times out of 10. The ANN provides the most robust Class 0 Recall (0.76 for Adoption and Default), making it the most reliable model identifying safe, low-risk users. Detailed neural network loss curves and AUC performance metrics are provided in [Fig. 13](#) to complement the confusion matrix results.

Models performance table								
Metrics	BNPL Adoption				BNPL Default			
	ANN	Logistic	SVC	XGBoost	ANN	Logistic	SVC	XGBoost
ROC-AUC	0.77	0.77	0.78	0.76	0.74	0.76	0.77	0.67
Class0 Precision	0.92	0.91	0.95	0.94	0.90	0.91	0.92	0.90
Class0 Recall	0.76	0.70	0.72	0.77	0.76	0.72	0.73	0.63
Class1 Precison	0.25	0.34	0.03	0.26	0.34	0.34	0.34	0.32
Class1 Recall	0.62	0.69	0.68	0.62	0.60	0.66	0.65	0.71

Table. 1. A comparison between the 4 models

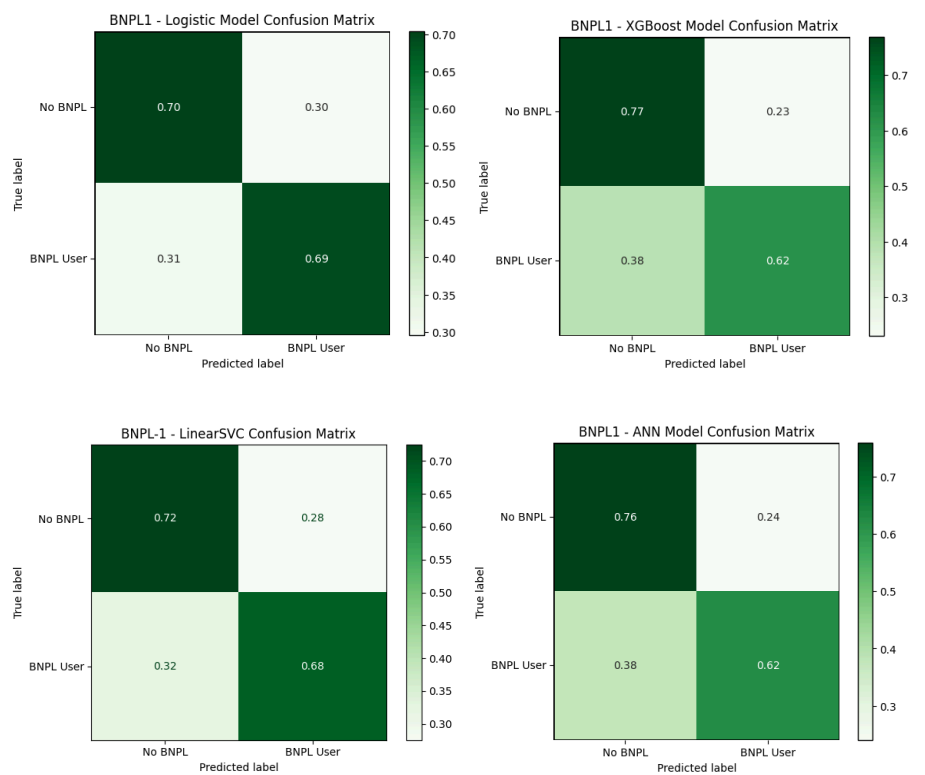
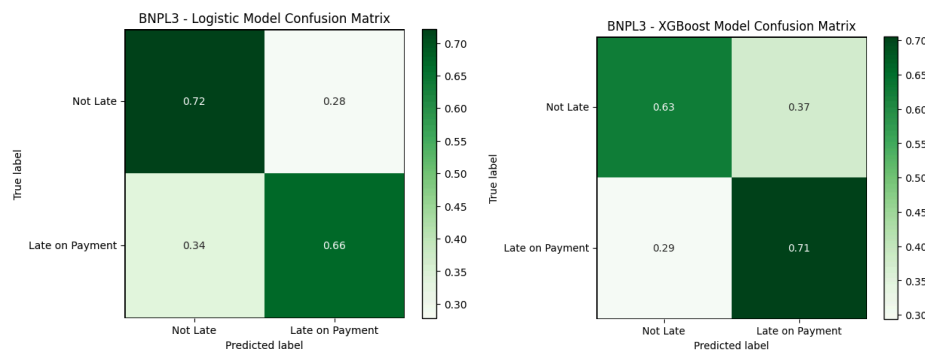


Fig. 11. BNPL adoption confusion rates across 4 models: Logistic, XGBoost, SVC, and ANN



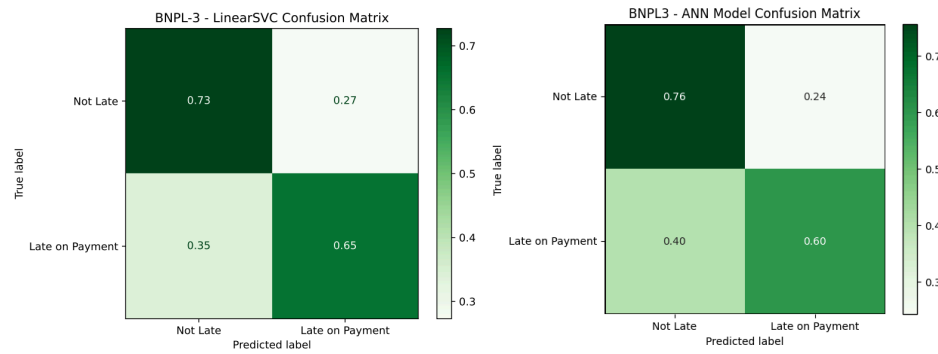


Fig. 12. BNPL delinquency confusion rates across 4 models: Logistic, XGBoost, SVC, and ANN

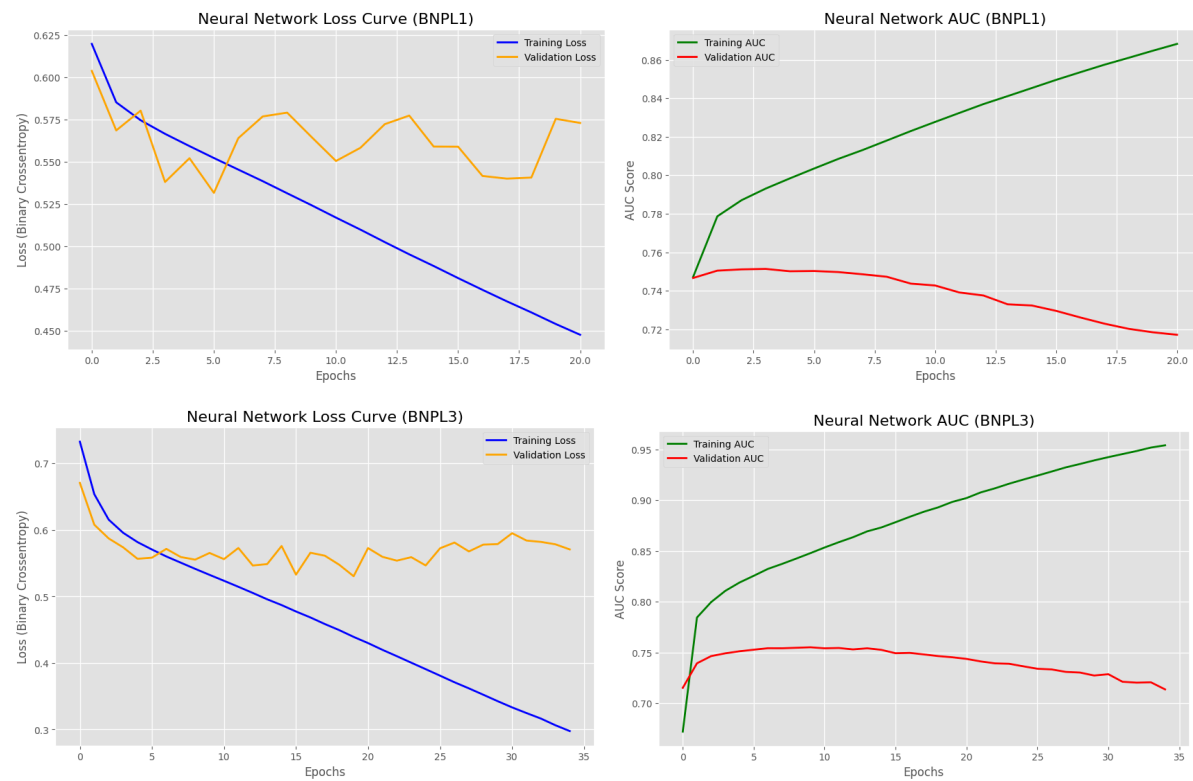


Fig. 13. ANN Model Loss and AUC curves

4.2. Feature Importance and Gain Analysis

XGBoost gain plots ([Fig. 14-15](#)) reveal that while both models share several high-impact features, they diverge in their primary drivers for adoption versus default behaviors.

- Distinct Primary Predictors:** The models show a divergence at the top of the hierarchy. BNPL Adoption is overwhelmingly driven by C4A (Frequency of carrying an unpaid

balance), which holds a gain score of over 100—more than double any other feature. In contrast, BNPL Default is primarily driven by A6 (Credit card approval confidence).

- **Feature Overlap (8/15):** Despite the different rankings, the models share 8 core features: A6, ppfs1482, EF3_c, EF3_f, ppethm, ppage, ppfs0596, and EF1. The presence of these variables in both lists confirms that financial health and liquid asset availability are foundational to both the decision to use BNPL and the risk of default.
- **Model-Specific Insights:** Adoption (BNPL1) weights heavily on habit-based markers like setting emergency funds and carrying unpaid balance. Default (BNPL3) focuses more on economic capacity and capability, highlighting marital status, household income, and credit health. However, the overlapping 8 features suggests that financial fragility connects the two groups.

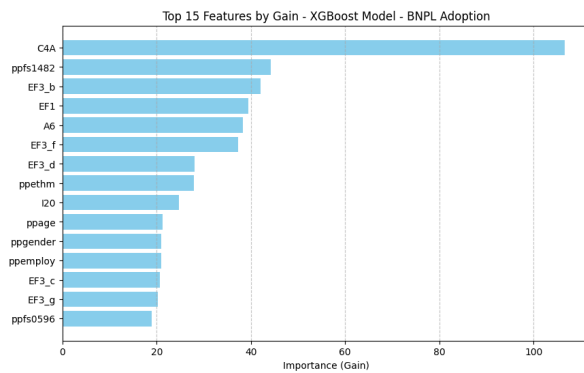


Fig. 14. BNPL adoption top 15 features by gain from XGBoost Model

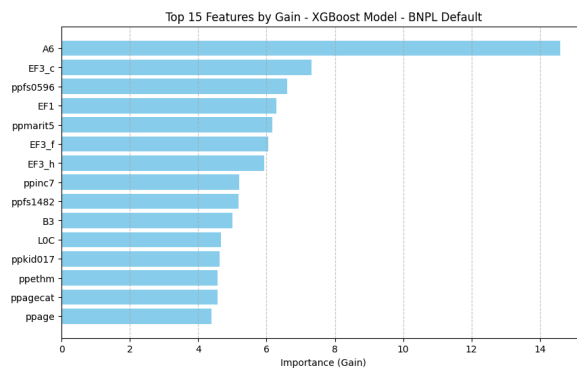


Fig. 15. BNPL delinquency top 15 features by gain from XGBoost Model

4.3. SHAP Global Interpretability

To provide a post-hoc interpretation of the model's internal logic and feature contributions, we utilized **SHapley Additive exPlanations (SHAP)** to quantify the global attribution of each predictor toward the final classification. The results are illustrated in [Fig. 16](#). SHAP values reveal the direction and the strength of influence of the top features on the XGBoost model's output.

- C4A (Frequency of carrying unpaid balance):** The strongest negative predictor for BNPL Adoption is C4A_Never carried an unpaid balance. High values (Red) for this feature correlate with significant negative SHAP impacts towards prediction, confirming that "full-payers" are the least likely to use BNPL.
- A6 (Credit card approval confidence):** In the Default model, the higher value of **A6_Very confident** pushes the SHAP value lower, indicating a factor against delinquency. Conversely, lower age (ppage) pushes SHAP values higher, suggesting that younger individuals are more likely to be BNPL users and potential defaulters.
- I20 (Spending vs. Income) & EF3 (Ways to pay \$400 emergency expense) :** A high spend - income ratio is correlated with no adoption of BNPL, while users who feel their spending is mismatched with their income are pushed toward higher probability towards higher risk/adoption.
- EF3_c (Pay with checking/saving/cash) & EF3_f (Pay with loans):** EF3_c shows a strong negative SHAP impact on both adoption and delinquency, suggesting that users who primarily utilize liquid assets are significantly less likely to rely on BNPL services or face payment defaults. Conversely, the feature EF3_f acts as a positive predictor for both models. High reliance on loans correlates with an increased probability of BNPL adoption and a higher risk of BNPL3 delinquency.

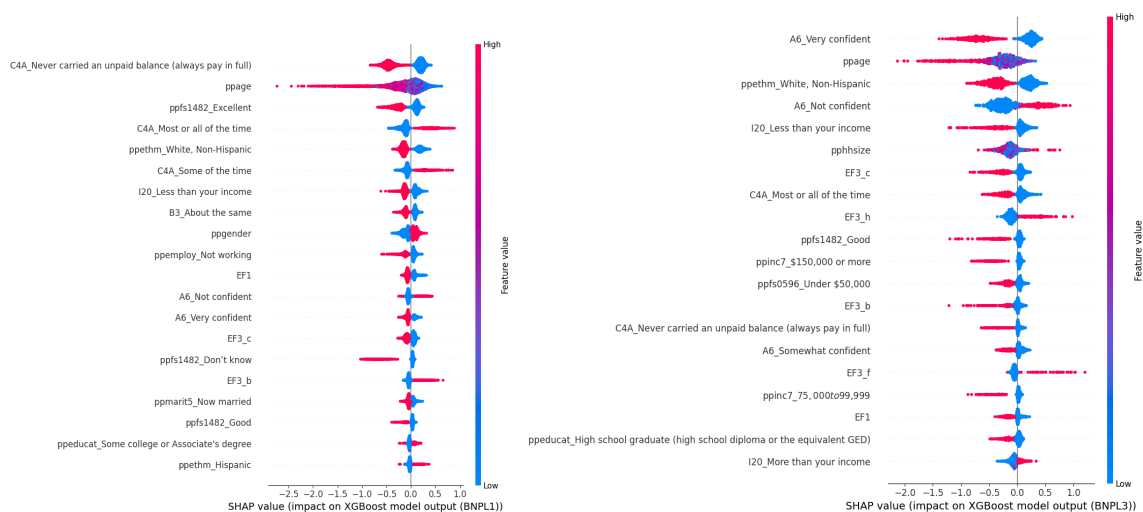


Fig. 16. SHAP value plots for BNPL adoption and BNPL delinquency

5. Discussion & Conclusion

5.1. Exploratory analyses

Buy Now, Pay Later services are often advertised as flexible, interest-free payment tools, helpful for managing short-term spending. However, after reviewing the literature, conducting descriptive analysis, and comparing multiple machine learning models, we observed that frequent users of BNPL services experience long-term financial vulnerabilities, and that BNPL use may contribute to users' struggles, rather than alleviate them.

We found BNPL adoption is strongly connected to financial stress. People who frequently carried unpaid credit card balances were much more likely to use BNPL than those who always paid their balances in full. Younger consumers, especially those between 25 and 34 years old, also showed the highest adoption rates, while older groups used BNPL much less. Household savings followed the same pattern: people with lower savings were much more likely to adopt BNPL, while people with stronger financial reserves showed much lower usage. These findings suggest that BNPL is often used as a short-term coping tool rather than simply a payment preference. This complements the previously discussed "flypaper effect," where BNPL creates vicious cycles of increased spending, overdrafts, and late fees.

We observed similar patterns in regards to BNPL loan repayments; lower-income users had much higher late payment rates than higher-income users, and users who were unconfident about being approved for credit were also more likely to miss BNPL payments. Users with fewer household savings and emergency funds showed similar repayment problems. Thus, defaulting behavior is not necessarily the result of a user's irresponsible spending, but structural financial vulnerabilities and weaker access to traditional credit systems.

BNPL use is also connected to family responsibility, especially for women with children. Women with children were significantly more likely to adopt BNPL than women without children, and this effect became stronger as the number of children increased. Even after controlling for income level and marital status, mothers consistently showed higher BNPL adoption rates. This suggests that BNPL is not only related to income differences, but also to caregiving responsibilities and household consumption pressure. Childcare, food, school expenses, and medical costs create regular financial demands, which may increase the need for installment-based payments. This helps us understand BNPL not only as a financial issue, but also as a family and social issue.

5.2. Model Performance

Across Logistic Regression, SVC, XGBoost, and Neural Network models, variables related to financial condition, such as income level, unpaid credit card balances, household savings, emergency fund access, and confidence in credit card approval repeatedly appeared as the most important predictors. Both the XGBoost feature importance results and SHAP

interpretation showed that financial fragility is the strongest common factor behind both BNPL adoption and the tendency to default on a BNPL loan.

SVC achieved the strongest overall ROC-AUC performance for both BNPL adoption and late payment prediction. However, XGBoost performed better in Class 1 Recall for BNPL default prediction, meaning it was better at identifying users who were actually at risk of missing payments. Practically speaking, this is more important to policymakers and financial institutions than pure accuracy, as missing a high-risk user can be more harmful than making a small classification error.

The neural network underperformed against simpler models and showed clear signs of overfitting. Logistic Regression, SVC, and XGBoost were easier to interpret and often produced more stable and reliable results.

5.3. Challenges and limitations

The primary challenge encountered in this study was the high dimensionality of the original dataset, which comprised over 100 survey questions. To mitigate the risk of overfitting and to isolate the most influential predictors, we began with a comprehensive Exploratory Data Analysis (EDA) to evaluate the strengths of individual relationships, followed by Principal Data Analysis (PCA). By utilizing these statistical techniques, we were able to reduce noise and pinpoint the underlying factors that exert the most significant influence on the predictive variables. A secondary challenge involved the inconsistency of the survey structures across the years, specifically, categorical notations shifted between 2021 and 2024. To ensure cohesive analysis, we conducted rigorous data harmonization and preprocessing, including standardizing disparate variables.

A notable limitation of this research is the reliance on the self-reported data concerning participants’ financial situations and concerns rather than objective credit records or bank data. This sentiment data are subject to recall and recognition bias, thus impacting the predictive results. Consequently, sourcing data directly from banking institutions, credit bureaus, or BNPL providers would be a more objective and real-time reflection of consumer behavior.

5.4. Final remarks

For many financially pressured consumers, BNPL functions as a coping mechanism for everyday financial pressure, and can worsen a user’s financial straits over time. Instead of only asking whether people are using BNPL responsibly, we should also ask why so many households need BNPL just to manage normal living expenses. The rapid growth of BNPL may reflect not only changes in payment technology, but also deeper problems of financial insecurity in daily life. Future researchers should explore the unique demographic characteristics of BNPL users we’ve identified, particularly among mothers and users from large families.

Appendix: Data Dictionary for Relevant Features

A6	If you were to apply for a credit card today, how confident are you that your application would be approved?
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B3	Whether respondent's family is better off financially at present, compared to the previous month
C4A	Number of times respondent carried an unpaid balance in the past 12 months
EF1	Have you set aside emergency or rainy day funds that would cover your expenses for 3 months in case of sickness, job loss, economic downturn, or other emergencies?
EF3_c	Q: If you had an emergency expense that cost \$400/\$500, how would you pay for the expense? A: With the money currently in my checking/savings account or with cash.
EF3_d	Q: If you had an emergency expense that cost \$400/\$500, how would you pay for the expense? A: With a bank loan or credit line
EF3_f	Q: If you had an emergency expense that cost \$400/\$500, how would you pay for the expense? A: With a payday loan, deposit advance, or overdraft
EF3_h	Q: I wouldn't be able to pay for the expense right now. A: With a payday loan, deposit advance, or overdraft
I20	Whether or not household spending was greater than or less than household income
L0C	Number of children in the household under the age of 18
ppage	Respondent age
ppagecat	Age (represented in categorical ranges, e.g. 18-24, 25-34)
ppemploy	Employment status
ppethm	Race/ethnicity
ppfs0596	Approximate total amount of investments and savings
ppfs1482	Where do you think your credit score falls?
ppgender	Respondent gender
ppinc7	Household income
ppkid017	Presence of kids in the household
ppmarit5	Marital status

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